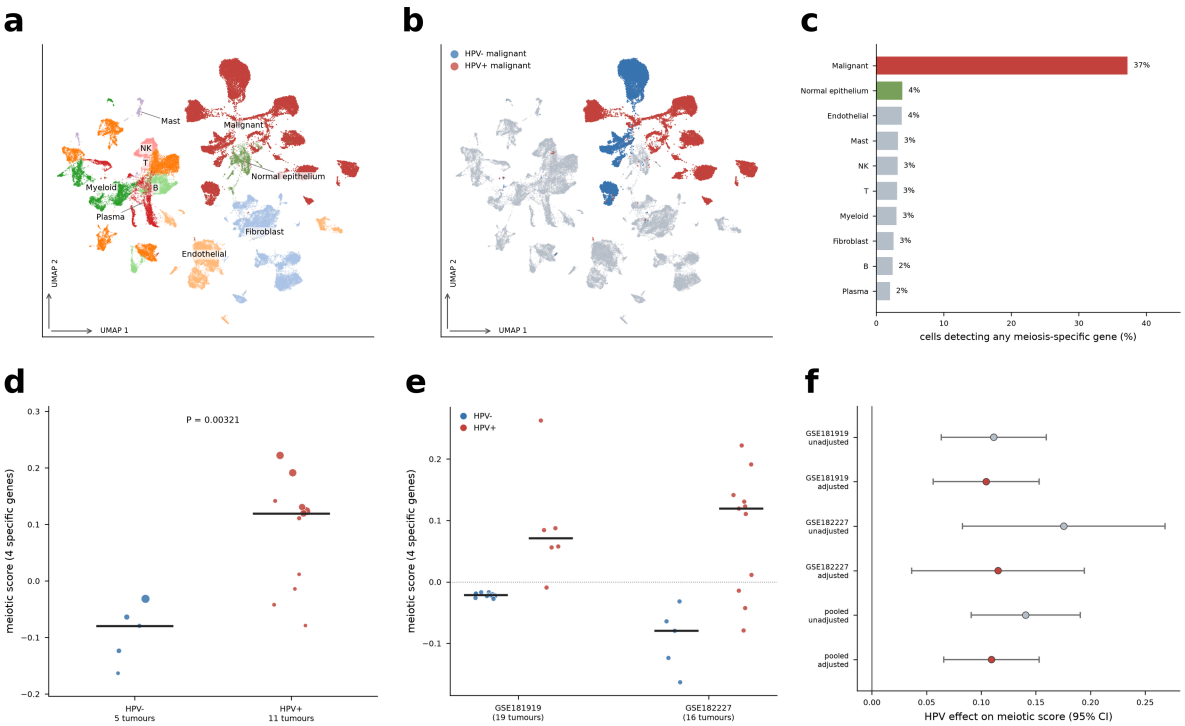

a
Mast
Malignant
NK
T
Myeloid
B
Normal epithelium
Plasma
Fibroblast
Endothelial
UMAP 2
UMAP 1
b
HPV- malignant
HPV+ malignant
UMAP 2
UMAP 1
c
Malignant
Normal epithelium
Endothelial
Mast
NK
T
Myeloid
Fibroblast
B
Plasma
37%
4%
4%
3%
3%
3%
3%
3%
2%
2%
0
10
20
30
40
cells detecting any meiosis-specific gene (%)
d
P = 0.00321
meiotic score (4 specific genes)
0.3
0.2
0.1
0.0
−0.1
−0.2
HPV-
5 tumours
HPV+
11 tumours
e
HPV-
HPV+
meiotic score (4 specific genes)
0.2
0.1
0.0
−0.1
GSE181919
(19 tumours)
GSE182227
(16 tumours)
f
GSE181919
unadjusted
GSE181919
adjusted
GSE182227
unadjusted
GSE182227
adjusted
pooled
unadjusted
pooled
adjusted
0.00
0.05
0.10
0.15
0.20
0.25
HPV effect on meiotic score (95% CI)